

Frame Analysis Lab

Luke Fountain

January 20th, 2026

Overview: This lab focused on performing manual frame and packet analysis using hexadecimal packet dumps captured from network traffic. The objective of the lab was to identify and interpret the different fields contained within Ethernet, IPv4, and TCP headers in order to better understand how network communication occurs at the packet level.

Question 1: The following is a complete Ethernet frame that was captured using TCPdump:

```
001E C983 E4F7 E046 9A8D 5736 0800 4500
003C B957 0000 3906 D8A9 4A7D E297 C0A8
01FE 0050 E51D E4B2 E268 260A EFDB A012
A62C 2E50 0000 0204 0596 0402 080A 22CF
6569 09A5 2F8A 0103 0306
```

Identify each of the values from the frame hex dump and put them into the table:

Source Mac Address	E046 9A8D 5736 or E0:46:9A:8D:57:36
Destination Mac Address	001E C983 E4F7 or 00:1E:C9:83:E4:F7
Payload	0800 or IP packet
Source IP	4A7D E297 or 74.125.226.151
Destination IP	C0A8 01FE or 192.168.1.254
IP Header Length	20 bytes
Protocol type	06 or TCP
Time to Live value	57
Source port	80
Destination port	58653
Sequence number	E4B2 E268 or 3,836,928,616
TCP flags	Ack flag

Packet Type	TCP packet with ACK flag set, likely the final acknowledgment in the three-handshake between a server on port 80 and a client on ephemeral port 58653
-------------	---

Question 2: Examine this IPv4 packet hex dump and answer the questions below.

```

45 10 00 36 32 3f 40 00 40 06 50 b7 c0 a8 66 83
8e 37 02 59 bc b6 00 15 b7 22 d9 58 36 25 63 7f
50 18 72 10 7c b1 00 00 50 41 53 53 20 6d 72 72
6f 62 6f 74 0d 0a

```

Identify each of the values from the frame hex dump and put them into the table:

IP header length	20 bytes
TCP header length	20 bytes
Payload length	14 bytes
Source IP address	192.168.102.131
Destination IP address	142.55.2.89
Protocol type	TCP
Time to Live	64
Source port	48246
Destination port	21
TCP flags	ACK and PSH
Payload type	Mr robot
Packet type?	TCP packet with ACK and PSH flags set, likely used in pushing an urgent packet from a client to a FTP server on port 21